Band-1 Inverse Galois: One Finite Simple Group over Q — A5 Probe (Collapse 2.32)

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Band-1 Inverse Galois — One Group (A_5) Probe

Honesty stamp. This note does **not** claim the Inverse Galois Problem solved. It does **not** claim that every finite group (or every finite simple group) is a Galois group over $\mathbb{Q}$. It does **not** claim that this probe realized a **new** finite simple group. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope.

> **Group chosen:** $A_5 \cong \mathrm{PSL}_2(\mathbb{F}_5)$ (order 60) — **classical** realization over $\mathbb{Q}$.
> **Full Inverse Galois over $\mathbb{Q}$: OPEN** (Band 4 / H-4.16).
> **Collapse 2.32** = one-group incremental ledger, **not** full IGP.

Abstract

The Inverse Galois Problem asks whether every finite group G arises as $\mathrm{Gal}(K/\mathbb{Q})$ for some Galois number field $K/\mathbb{Q}$. **Full IGP remains open.** Many individual finite simple groups are known over $\mathbb{Q}$ by a century of incremental work (Hilbert, rigidity, modular/automorphic images, Hilbert modular surfaces, ...). This Band-1 probe locks the status of **one** classical finite simple group — A_5 — with an optional on-box Dedekind / discriminant sanity check for the classical quintic

$$f(x) = x^5 + 20x + 16 \in \mathbb{Z}[x].$$

Box confirms: f irreducible over $\mathbb{Q}$; $\mathrm{disc}(f) = 32000^2$ (square) $\Rightarrow \mathrm{Gal}(f/\mathbb{Q}) \leq A_5$; mod-3 type (5) and mod-7 type $(1,1,3)$ force a 5-cycle and a 3-cycle, hence $\mathrm{Gal}(f/\mathbb{Q}) = A_5$. **This is a classical known fact, re-checked — not a new theorem.**

Problem (collapse 2.32 → Band-1 lit + one witness)

1. Pick **one** finite simple group with careful ledger (classical or recently realized).
2. Optional tiny $\mathrm{Gal}(f/\mathbb{Q})$ computation for an A_n/S_n witness.
3. Keep Clay / ToE / GATE-04 / "Inverse Galois solved" off the claim surface.
4. State the **open remainder**.

Literature ledger (web check; not re-proved on box)

Year / era	Result	Authors / source	Status for this probe
Hilbert classical	S_n, A_n over $\mathbb{Q}$ (and regularly over $\mathbb{Q}(t)$)	Hilbert; many explicit families	A_5 **classical known**

Year / era	Result	Authors / source	Status for this probe
—	Smallest nonabelian simple = A_5	—	first nonsolvable IGP witness class
rigidity era	Many Lie-type over prime fields; most sporadics	Malle–Matzat; others	secondary context
Zywina lists	Catalog of small simple groups not yet known over $\mathbb{Q}$	Zywina	inventory tool
2015+	$\mathrm{PSL}_2(\mathbb{F}_p)$ for all p (non-regular)	Zywina	lit known (not A_5)
2022	Seven small simple groups newly realized over $\mathbb{Q}$ (e.g. $\mathrm{PSL}_2(\mathbb{F}_{3^3})$, $\mathrm{PSU}_3(\mathbb{F}_5)$, $\mathrm{PSL}_3(\mathbb{F}_7)$, ...)	Dieulefait–Taixés et al. (MDPI)	**other** groups; not claimed here
recent	All 26 sporadics known over $\mathbb{Q}$; M_{23} regularly over $\mathbb{Q}(t)$	lit (MM / follow-ups)	secondary; not the chosen group
arXiv:2512.23615	Regular $\mathrm{PSL}_2(\mathbb{F}_{p^2})$ under Kronecker conditions on a finite set $\mathcal{L}$ of primes ≤ 41 ($\neq 31$)	Demeio–Gvirtz-Chen	**family progress**; congruence gaps remain
arXiv:2604.27402	Claims explicit infinite $\mathrm{PSL}(n,q)/\mathrm{PSU}(n,q)$ series with growing rank and field degree	lit preprint	**CLAIM/lit only** — not re-proved; not elevated as house-closed
—	**Full Inverse Galois over** $\mathbb{Q}$	—	**OPEN** (Band 4)

Canonical classical witness. $f = x^5 + 20x + 16$: discriminant square $\Rightarrow$ Gal $\leq A_5$; Dedekind reductions supply a 5-cycle and a 3-cycle; subgroup index constraints force equality with A_5 (standard textbook argument). Equivalent translate $x^5 - 5x^4 + 10x^3 - 10x^2 + 25x - 5$ (Eisenstein at 5) appears in formalizations.

Why A_5 for this probe. Collapse 2.32 asks for an honest **one-group** ledger, not a dash invention of a new realization. $A_5 \cong \mathrm{PSL}_2(\mathbb{F}_5)$ is the cleanest classical finite simple group with an elementary explicit polynomial and a reproducible box check. Claiming a "new" group here would be dishonest.

What is proved vs open (statement lock)

Known for the chosen group (literature + box sanity)

1. A_5 **is a Galois group over** $\mathbb{Q}$ — classical.
2. A_n **for all** $n \geq 5$ and S_n **for all** n — classical over $\mathbb{Q}$.
3. **Box re-check** of one witness polynomial (below) — computational sanity, not a new proof.

Locked OPEN (remainder)

1. **Full Inverse Galois Problem:** every finite group G arises as $\mathrm{Gal}(K/\mathbb{Q})$ — **OPEN** (Band 4 / H-4.16 / NT-INVGAL).
2. **Regular Inverse Galois** for many Lie-type groups over composite $\mathbb{F}_q$ — incomplete; congruence / unirationality gaps remain (e.g. $\mathrm{PSL}_2(\mathbb{F}_{p^2})$ without the Demeio–Gvirtz-Chen Kronecker conditions).
3. **Large classes of** $\mathrm{PSL}(n,q)$**,** $\mathrm{PSU}(n,q)$**, exceptional groups of Lie type** — still open or only partially known; 2026 preprint families are **lit/CLAIM**, not re-proved here.
4. **This probe does not add a new simple group to the known list.**

Optional box illustration (not a new theorem)

Script `probe_a5_galois.py` (sympy) on $f = x^5 + 20x + 16$:

Check	Result
Irreducible over $\mathbb{Q}$?	**True**
$\mathrm{disc}(f)$	$1{,}024{,}000{,}000 = 32000^2$ (**square**) $\Rightarrow$ Gal $\leq A_5$
Factor type mod 3	$(5) \Rightarrow$ contains a 5-cycle
Factor type mod 7	$(1,1,3) \Rightarrow$ contains a 3-cycle
Conclusion (classical)	$\mathrm{Gal}(f/\mathbb{Q}) = A_5$

S_5 **control** $g = x^5 - 4x + 2$: irreducible; mod-5 type $(1,2,2)$ supplies an odd permutation (double transposition), consistent with Gal $= S_5$ (standard), contrasting the even A_5 witness.

Label: Dedekind / discriminant sanity only. **Not** a search for unknown groups. **Not** a solution of Inverse Galois.

Why Band-1 vs conjecture

- **Band-1 (this probe):** honest one-group ledger for classical A_5; optional Gal witness.
- **Band 4:** full Inverse Galois over $\mathbb{Q}$ for all finite groups.
- **Not claimed:** IGP solved; all simple groups settled; new realization by ZoraASI; Clay; ToE.
- **GATE-04:** LOCKED.

Open remainder (ship line)

Group chosen: $A_5 \cong \mathrm{PSL}_2(\mathbb{F}_5)$ — classical **REALIZED** over $\mathbb{Q}$ (box re-check PASS).
Open remainder: full Inverse Galois (all finite G) **OPEN**; many Lie-type / regular cases still open or congruence-conditional; this probe adds **no new** simple group.